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Marc Andreessen and the Trap of Technological Determinism

Marc Andreessen argues that artificial intelligence will significantly improve various aspects of society in his piece “Why AI Will Save the World.” He covers a wide variety of applications under the umbrella of “artificial intelligence,” but I will concern myself specifically with the applications of large language models (LLMs). These models generate and process human-like text by being trained on large amounts of text data to perceive statistical relationships and language patterns. LLMs can perform a variety of tasks, including answering questions, writing, and translating. Although Andreessen effectively refutes more alarming LLM fears, his failure to quell nuanced anxieties through his tech-hype narrative provides a useful map to the weaknesses of technological determinism. In this essay, I seek to deconstruct Andreessen’s argument and investigate the narratives motivating it, as well as offer an objection to his technologically determinist perspective. First, I will define the viewpoint of technological determinism and explain how Andreessen’s opinion can be categorized as such. Then, I will introduce the idea of co-production between technology and society and explain how this perspective invalidates determinism by emphasizing the impact that the dominant socio-economic paradigms have on the construction and use of LLMs. Finally, I will end by iterating over a few key ethical risks posed by LLMs that Andreessen overlooks.

Technological determinism describes the idea that technology shapes society around it, and changes in technology influence changes in society. These changes are assumed to be seen in

behavior, institutions, and values. Technological determinism is less of a defined school of thought and more of a description of an ideology held by people concerning technology. A sentiment commonly associated with it is that technology evolves in a realm independent of cultural or societal influence and develops with a logic on its own. Andreessen echoes this sentiment by reasoning that "human intelligence makes a very broad range of life outcomes better," and thus "anything that people do with their natural intelligence today can be done much better with AI […] from curing all diseases to achieving interstellar travel." Through this line of reasoning, Andreessen takes the position that the eventual development of AI is a certainty due to intelligence being the mover of innovation. Andreessen further reiterates his deterministic narrative by stressing "[h]istorically, every new technology that matters, from electric lighting to automobiles to radio to the Internet, has sparked a moral panic[.]" Andreessen attributes the societal response as the effect of technology, where the technology creates new moral dilemmas as part of innovation rather than being symptoms of persevering moral choices in dialectic with themselves.

Co-production objects to technological determinism by asserting a bidirectional relationship between technology and society, a relationship that implies a mutual dependence for the forms that they both take. For example, the adoption of automobiles in rural American communities contrasts Andreessen's deterministic interpretations by agricultural users of the technology customizing and shaping cars to fit their own needs, not simply passively accepting what manufacturers offered them. This resulted in models being manufactured to fit those needs, changing the automobile industry (Kline, Pinch). This is not to say that automobiles and other technology did not result in new moral dilemmas, but Andreessen missteps by wholly separating technology from users. He continues this trend through the broad predictions he makes about

potential LLM uses, asserting "[w]hat AI offers us is the opportunity to profoundly augment human intelligence to make all of these outcomes of intelligence." Andreessen lists scientific AI collaborators, AI tutors, and even the creative arts entering a "golden age" as a result of this AI augmentation. In accordance with his previous suppositions, Andreessen misses the impact that users have on the creation of technology. Ironically enough, Andreessen is calling for a culture shift in order to embrace AI while at the same time avoiding the cultural conditions that created it. The decision-making paradigm introduced by 19th and 20th century researchers laid the groundwork for LLM model training. The psychedelics researchers of the early 1900's were the first to consider the brain as a machine that could be modeled and manipulated with chemical keys. And most significantly, the capitalist economic structure of labor and profit spurred the pursuit of automation and replacing expensive human labor, a practice that has been accumulating since the start of the Industrial Revolution. Andreessen addresses through an underhanded argument, stating that "When technology is applied to production, we get productivity growth – an increase in output generated by a reduction in inputs. The result is lower prices for goods and services." While these highly normative straw-man arguments of technological determinism and Keynesian economics look good on paper, the reality is much more complicated. Companies rarely lower prices for consumers, unless required to through competition or regulation. Furthermore, Andreessen's claim that "technology empowers people to be more productive" similarly does not hold in all cases. Some technology, such as the self-checkout kiosks mentioned by Andreessen, simply allows for companies to cut workers and overhead, making the company more productive but not necessarily individual workers. Much like the technical upheavals of the past, reskilling labor and the lagging emergence of new industries will leave many workers stranded.

While Andreessen attempts to soothe the most prominent AI fears of economic disparity (which he misses the mark on) and a total robot apocalypse (which he does well, but the doomsayers have been wrong for all of human existence), he fails to address why regulation would be a bad move for mitigating the ethical risks of AI. Most actions are illegal no matter who does it: racial discrimination is the same whether a human reads a résumé or an LLM; harmful disinformation, although more difficult to prosecute because of the highly politicized definition of the word, is illegal no matter the actor. In most cases, and I would say the same for LLMs, the ability to create industry standards (such as those that revolutionized airline safety) is an easier regulatory step that should be pursued. And finally, between the two positions of total apocalypse and material utopia presented by Andreessen, I believe that the resulting world will be somewhere between the two.

Works Cited

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Kline, Ronald, and Trevor Pinch. 1996. "Users as agents of technological change: The social construction of the automobile in the rural United States." Technology and culture 37 (4): 763-795.